

Stoke's Theorem (Relation b/w line and surface integrals)

→ Let S be an open surface bdd by a closed curve C and $\vec{F} = F_1 i + F_2 j + F_3 k$ be any vector pt fn having contin first order partial derivatives. Then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, dS, \text{ where } \hat{n} \text{ is a unit}$$

vector at any point of S drawn in the sense in which

Q. Verify Stoke's thm for $\vec{F} = (x^2 + y^2)i - 2xyj$ taken around the rectangle bdd by the lines $x = \pm a, y = 0, y = b$?

Let C denote the bdr of the rectangle $ABED$. Then

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \oint_C (x^2 + y^2)i - (2xyj) \cdot (i dx + j dy) \\ &= \oint_C (x^2 + y^2) dx - 2xy dy \end{aligned}$$

The curve C consists of 4 lines AB, BE, ED and DA.

Along AB, $x=a$, $dx=0$, $0 \leq y \leq b$.

$$\therefore \int_{AB} (x^2 + y^2) dx - 2xy dy = \int_0^b -2ay dy = -ay^2 \Big|_0^b \\ = \underline{\underline{-ab^2}}$$

Along BE, $y=b$, $dy=0$, $a \leq x \leq -a$

$$\therefore \int_{BE} (x^2 + y^2) dx - 2xy dy = \int_a^{-a} (x^2 + b^2) dx \\ = \left[\frac{x^3}{3} + b^2 x \right]_a^{-a}$$

$$= -\frac{a^3}{3} - ab^2 - \left[\frac{a^3}{3} - ab^2 \right] \\ = \underline{\underline{-\frac{2a^3}{3} - 2ab^2}}$$

Along ED, $x=-a$, $dx=0$ and $b \leq y \leq 0$.

$$\therefore \int_{ED} (x^2 + y^2) dx - 2xy dy = \int_b^0 -2ay dy = -ay^2 \Big|_b^0 \\ = \underline{\underline{-ab^2}}$$

Along DA, $y=0$, $dy=0$ and $-a \leq x \leq a$

$$\therefore \int_{DA} (x^2 + y^2) dx - 2xy dy = \int_{-a}^a x^2 dx = \left[\frac{x^3}{3} \right]_{-a}^a = \underline{\underline{\frac{2a^3}{3}}}$$

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = -ab^2 - \frac{2a^3}{3} - 2ab^2 - ab^2 + \frac{2a^3}{3} \\ = \underline{\underline{-4ab^2}} \quad \text{--- ①}$$

Now, $\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2+y^2 & -2xy & 0 \end{vmatrix} = (-2y-2y)\hat{k} = \underline{\underline{-4y\hat{k}}}$

For the surface S , $\hat{n} = \hat{k}$

$\therefore \text{curl } \vec{F} \cdot \hat{n} = -4y$

$$\iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds = \int_0^b \int_{-a}^a -4y \, dx \, dy$$

$$= \int_0^b -4y \, x \Big|_{-a}^a \, dy$$

$$= -4 \times 2a \int_0^b y \, dy$$

$$= -8a \cdot \left[\frac{y^2}{2} \right]_0^b$$

$$= \underline{\underline{-4ab^2}} \quad \text{--- (2)}$$

The equality of (1) and (2) verifies Stokes' thm.

Q. Verify Stokes' thm for the vector field $\vec{F} = (2x-y)\hat{i} - yz^2\hat{j} + yz^2\hat{k}$ over the upper half surface of $x^2+y^2+z^2=1$, bounded by its projection on the xy -plane?

Let S be the upper half surface of the sphere $x^2+y^2+z^2=1$. boundary C of S is a circle in the xy -plane of radius unity and centre O . The eqn of C are $x^2+y^2=1, z=0$ whose parametric form is, $x = \cos t, y = \sin t, z = 0, 0 \leq t \leq 2\pi$

$$\begin{aligned}
 \oint_C \vec{F} \cdot d\vec{r} &= \oint_C (2xy - y)i - yz^2j - y^2zk (i'dn + j'dy + k'dz) \\
 &= \oint_C (2x - y)dn - yz^2dy - y^2zdz \\
 &= \oint_C (2x - y)dn \quad (\because z=0, dz=0) \\
 &= \int_0^{2\pi} (2\cos t - \sin t) \frac{dn}{dt} dt \\
 &= \int_0^{2\pi} (2\cos t - \sin t) \cdot (-\sin t) dt \\
 &= \int_0^{2\pi} (-2\cos t \sin t + \sin^2 t) dt \\
 &= \int_0^{2\pi} \left(-\sin 2t + \frac{1 - \cos 2t}{2} \right) dt \\
 &= \left[\frac{\cos 2t}{2} + t/2 - \frac{\sin 2t}{4} \right]_0^{2\pi} \\
 &= \frac{1}{2} + \pi - \frac{0}{2} = \underline{\underline{\pi}} \quad \text{--- ①}
 \end{aligned}$$

Also $\text{curl } \vec{F} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2x - y & -yz^2 & -y^2z \end{vmatrix} = (-2yz + 2yz)i + 0j + 0k$

$$= \underline{\underline{\hat{k}}}$$

$$\text{curl } \vec{F} \cdot \hat{n} = \hat{k} \cdot \hat{n} = \hat{n} \cdot \hat{k}$$

$$\therefore \iint_S \text{curl } \vec{F} \cdot \hat{n} \, dS = \iint_S \hat{n} \cdot \hat{k} \, dS = \iint_R \hat{n} \cdot \hat{k} \frac{dxdy}{|\hat{n} \cdot \hat{k}|}$$

where R is the projection of S on xy -plane.

$$= \iint_R dx dy = \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy dx$$

$$= \int_{-1}^1 2\sqrt{1-x^2} dx$$

$$= 4 \int_0^1 \sqrt{1-x^2} dx$$

$$= 4 \left[\frac{x\sqrt{1-x^2}}{2} + \frac{1}{2} \sin^{-1}(x) \right]_0^1$$

$$= 4 \left[\frac{1}{2} \cdot \frac{\pi}{2} - 0 \right] = \underline{\underline{\pi}} \quad \text{--- (2)}$$

$$\therefore \textcircled{1} = \textcircled{2}.$$

Q. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by Stoke's thm, where $\vec{F} = y^2 \hat{i} + x^2 \hat{j} - (x+2) \hat{k}$ and C is the bdy of the Δ with vertices at $(0,0,0)$, $(1,0,0)$ and $(1,1,0)$.

Since z coordinate of each vertex of the Δ is zero, the Δ lies in the xy -plane and $\hat{n} = \hat{k}$.

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x^2 & -(x+2) \end{vmatrix}$$

$$= \underline{\underline{\hat{j} + 2(x-y) \hat{k}}}$$

$$\therefore \text{curl } \vec{F} \cdot \hat{n} = 2(x-y)$$

The eqn of line OB is $y=x$

By Stoke's thm,

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \iint_S \text{curl } \vec{F} \cdot \hat{n} \, dS \\ &= \int_0^1 \int_0^x 2(x-y) \, dy \, dx \\ &= 2 \int_0^1 \left[xy - \frac{y^2}{2} \right]_0^x \, dx \\ &= 2 \int_0^1 \left(x^2 - \frac{x^2}{2} \right) \, dx \\ &= 2 \int_0^1 \frac{x^2}{2} \, dx = \left[\frac{x^3}{3} \right]_0^1 = \underline{\underline{\frac{1}{3}}} \end{aligned}$$

Q. Evaluate the surface integral $\iint_S \text{curl } \vec{F} \cdot \hat{n} \, dS$ by transforming it into a line integral, S being that part of the surface of the paraboloid $z=1-x^2-y^2$, for which $z \geq 0$ and $\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$.

The bdy C of the surface S is the circle $x^2+y^2=1$, $z=0$ whose para eqns are, $x = \cos t$, $y = \sin t$, $z = 0$, $0 \leq t \leq 2\pi$.

By Stoke's thm, $\iint_S \text{curl } \vec{F} \cdot \hat{n} \, dS = \oint_C \vec{F} \cdot d\vec{r}$.

$$= \oint_C (y\vec{i} + z\vec{j} + x\vec{k}) \cdot (x\vec{i} + y\vec{j} + z\vec{k})$$

$$= \oint_C (y \, dx + z \, dy + x \, dz)$$

$$= \oint_C \psi \, dn \quad (\because \, z=0, \, dz=0).$$

$$= \oint_C \psi \cdot \frac{dn}{dt} \cdot dt = \int_0^{2\pi} \sin t \cdot (\sin t) \, dt$$

$$= - \int_0^{2\pi} \sin^2 t \, dt$$

$$= - \int_0^{2\pi} \frac{1 - \cos 2t}{2} \, dt$$

$$= -\frac{1}{2} \left[t - \frac{\sin 2t}{2} \right]_0^{2\pi}$$

$$= -\frac{1}{2} \left[2\pi + 0 \right]$$

$$= \underline{\underline{-\pi}}.$$